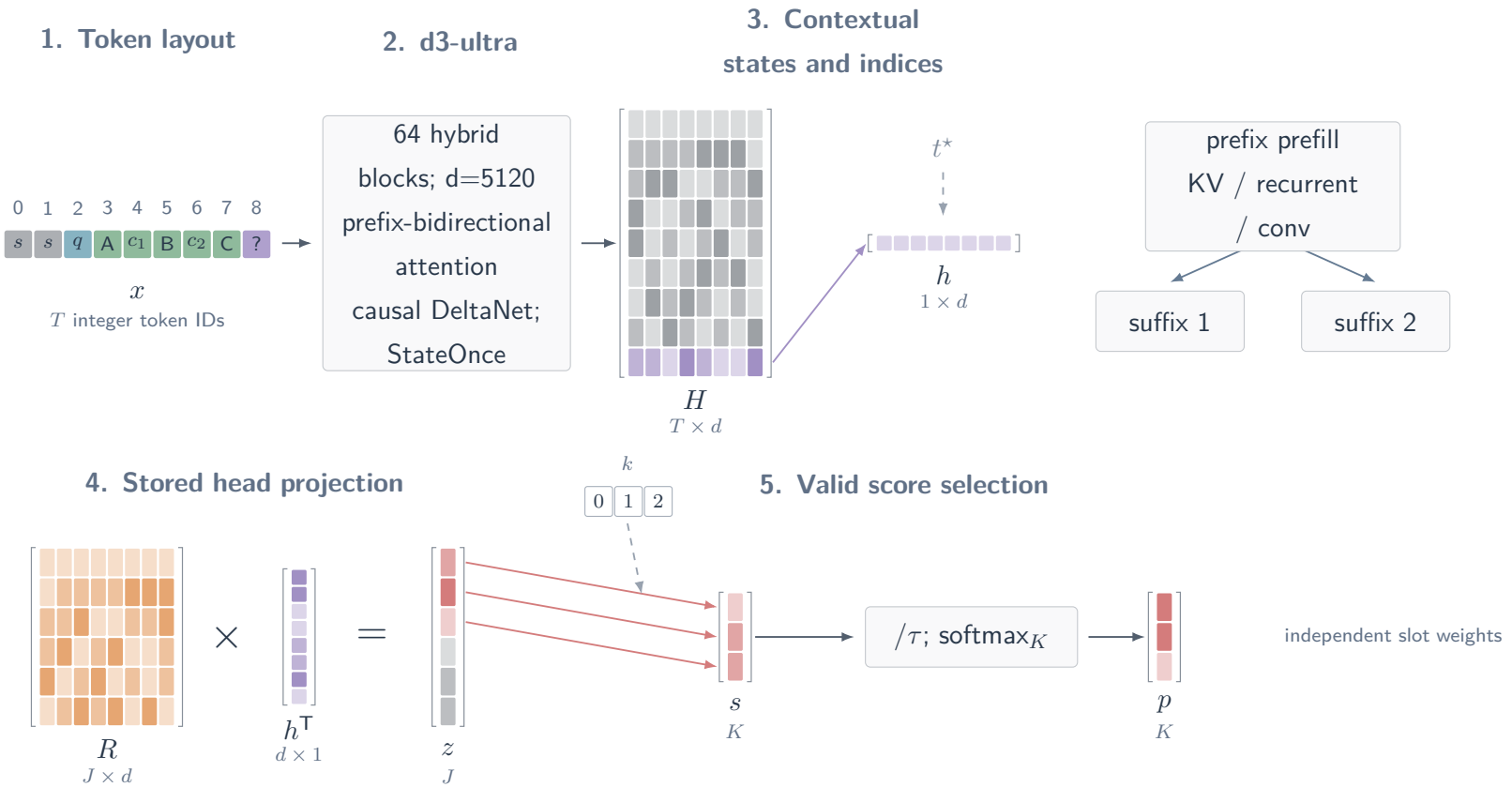
$$h = H_{t^\star,:}, \quad z = Rh^\mathsf{T}, \quad s = z_{0:K}, \quad p = \mathrm{softmax}_K(s/\tau)$$




| | |
|---|---|
| **Axes** | $d = 5120,\ J = 255$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes. |
| **Objects** | Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes. |
| **Mechanism** | Prefix prefill saves KV, recurrent and convolution state; each suffix forks that state. A suffix sees the prefix and its own text, never another suffix.<br>Independent 255 x 5120 readout; initial head and training objective undisclosed. The suffix-end hidden state is read after final normalization. |